

How an SD-SLO measures eye motion

A first-principles guide for readers who do not already know the instrument, its timing model, or the gaze-estimation pipeline

Status: 2026-07-22 • numerical claims are evidence-classified

What this document will explain

READING MAP

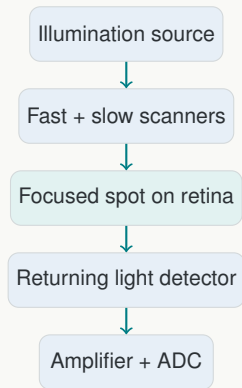
1. What an SLO is and how its optical/electronic components create a raster
2. Why a retinal image contains time, not only space
3. How strips and normalized correlation estimate motion
4. How pixel displacement becomes a calibrated gaze trajectory
5. What current synthetic and real-data results do and do not prove
6. Why inter-frame dead time matters for cognitive events
7. Which signals need temporal rate versus spatial precision
8. Which production interfaces should be frozen before hardware

How to read the evidence labels. “Measured” means observed on a stated capture; “derived” means arithmetic from stated inputs; “fixture” means synthetic known truth; “hypothesis” and “missing” explicitly mark claims that still require measurement.

Source: scope of this explanatory deck

An SLO is a point-scanning camera for the retina

HARDWARE PRIMER



Scanning laser ophthalmoscope (SLO). A small optical spot is moved across the retina. The detector records one intensity value at each instant. Scanner position and detector time together determine where that sample belongs in the retinal image.

Why this matters for gaze. If the eye moves while the raster is being acquired, vessels shift between early and late samples. Registration estimates that shift. The gaze signal is therefore derived from the time-dependent deformation of retinal structure, not from a separate pupil image.

Source: instrument-level interpretation of the technical note and repository scan model

Two scanner axes create one image

HARDWARE PRIMER + DERIVED

FAST AXIS

The resonant scanner sweeps the spot horizontally. At 10 kHz, one full sinusoid lasts 100 μs . Using both half-cycles yields a forward line and a reverse line, each 50 μs .

SLOW AXIS

A second scanner advances the beam vertically from line to line. A nominal frame is the software grouping of many sequential fast-axis lines.

$$T_{\text{sine}} = 1/f_{\text{scan}} = 100 \mu\text{s}$$

$$T_{\text{line}} = T_{\text{sine}}/2 = 50 \mu\text{s}$$

$$f_{\text{line}} = 2f_{\text{scan}} = 20,000 \text{ s}^{-1}$$

$$T_{808} = 808/f_{\text{line}} = 40.4 \text{ ms}$$

These numbers describe the candidate 10 kHz fast axis. They do not by themselves reveal slow-axis return time, optical blanking, or the duty cycle of the current capture.

Source: technical note sections 1–3; legacy 808-line geometry

The detector produces a time series before it produces an image

FIRST PRINCIPLES

Let detector samples be $I(t_k)$ and scanner position be $(x(t_k), y(t_k))$. Image formation assigns each sample to a retinal coordinate:

$$\{I(t_k), x(t_k), y(t_k)\}_{k=1}^N \longrightarrow I_{\text{rect}}(x_i, y_j).$$

The raw ADC is uniform in time. Because a sinusoid moves fastest at field center and stops at the turnarounds, those samples are *not* uniform in retinal space.

A FRAME IS A CONVENTION

A frame is a block of sequential lines chosen for image construction. The retina was not measured everywhere at one instant.

A MISSING INTERVAL IS PHYSICAL

If lines are not recorded during slow-axis return or blanking, no image-processing method can recreate the unobserved retinal photons.

Source: technical note sampling model; scan_timing.py; desinusoid.py

Eye motion is written into the raster as distortion

FIRST PRINCIPLES

Suppose a vessel has stationary retinal texture $R(x, y)$ and the eye displacement is $\mathbf{d}(t) = [d_x(t), d_y(t)]^T$. The detector approximately sees

$$I(t) \approx R(x_s(t) - d_x(t), y_s(t) - d_y(t)) + \eta(t),$$

where (x_s, y_s) is scanner position and η is detector/texture noise. Different parts of one frame therefore encode different eye positions.

Registration inverts this effect. A short strip spans a small time aperture. Matching that strip to a retinal reference estimates the displacement that best aligns the vessel pattern.

Trade-off. Wider strips contain more vessel structure and improve correlation, but average motion over a longer interval. Smaller stride emits estimates more often, but neighboring estimates reuse pixels.

Source: brightness-constancy motion model; strip registration implementation

The mission changed; the architecture mostly did not

PROGRAM FACT + ARCHITECTURE

2020 vector: home retinal health

- Portable SLO/OCT imaging
- Intermittent, high-quality retinal frames
- Patient monitoring between clinic visits
- Image quality and medical workflow dominate

Current vector: cognitive platform

- Continuous attention-aligned measurements
- Retina, pupil, stimulus, behavior, and OPM-MEG
- Hardware time, validity, uncertainty, and provenance
- Model-ready synchronized sequences

Keep the retinal capability; re-audit assumptions that let frame boundaries erase time.

Source: program history; OPM coexistence accepted as characterized

Every number carries an evidence class

EVIDENCE POLICY

MEASURED

Directly measured on the stated capture or hardware.

FIXTURE

Known-truth software or synthetic experiment.

MISSING

Required value absent; no numerical conclusion is permitted.

DERIVED

Calculated from cited inputs with an explicit formula.

PROGRAM FACT

Accepted project history or characterized capability.

LITERATURE

Reported by a primary or authoritative source.

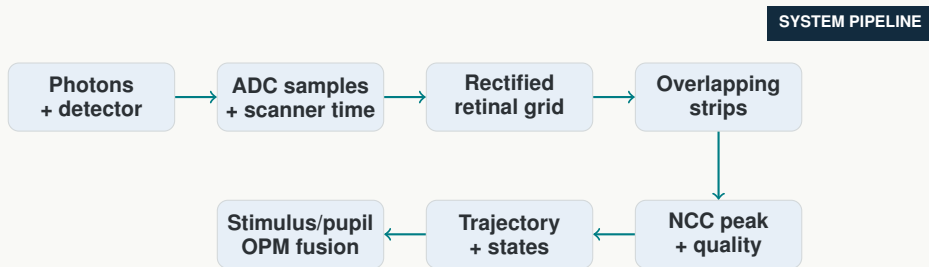
HYPOTHESIS

Candidate design or interpretation requiring validation.

- No synthetic-to-biological upgrade
- No prediction-to-observation upgrade
- No geometry-to-safety upgrade

Source: cognitive_gaze_evidence.json

From photons to an attention-aligned sample



Invariant record: acquisition time, direction, calibration version, quality, and observed/predicted/rejected/missing state.

Source: technical note; repository contracts; architecture synthesis

“Rate” is four different clocks

DERIVED

LINE CLOCK

20 kline/s

One 50 μ s half-sine per line

APERTURE CLOCK

1.20 ms

24 lines contribute to one estimate

UPDATE CLOCK

0.30 ms

6-line stride gives 3.33 kHz spacing

WALL CLOCK

Duty limited

Active updates stop in dead time

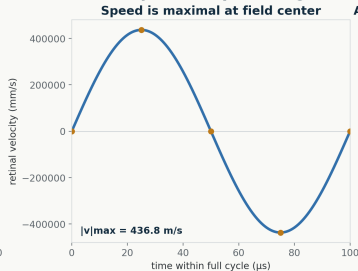
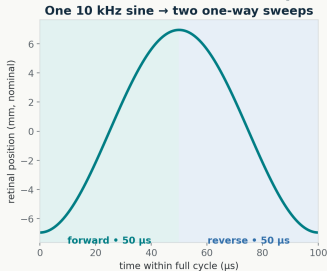
Overlapping estimates are correlated. Output spacing is not independent measurement bandwidth.

Source: technical note; 24/6 strip design example

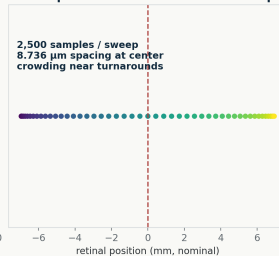
A 10 kHz sine is a 20 kline/s bidirectional clock

DERIVED

Fast-axis acquisition clock: every stored pixel begins as a timestamp



ADC samples are uniform in time—not space



Source: technical note sections 2–3 and 7

Angular field becomes millimeters through the eye model

HYPOTHESIS + DERIVED

$$\begin{aligned}q &= 0.01306(AL - 1.82) \\q(24 \text{ mm}) &= 0.2896708 \text{ mm/deg} \\W &= 48^\circ q = 13.9042 \text{ mm}\end{aligned}$$

Model boundary

The 24 mm axial length and 48° field are nominal design inputs. Subject-specific millimeters require measured biometry and calibrated scan geometry.

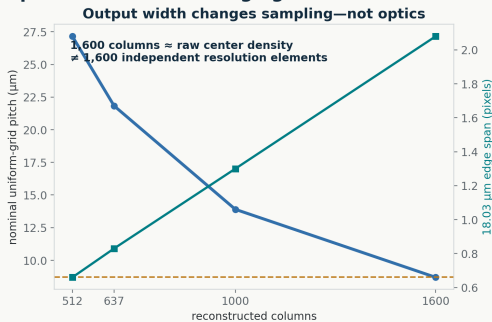
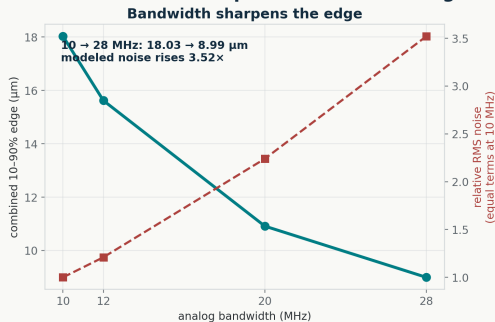
Pixel displacement is measured first; retinal angle or millimeters require explicit calibration.

Source: technical note sections 1–2; Bennett et al. 1994

Bandwidth, optics, and stored pixels are not interchangeable

MEASURED + DERIVED + HYPOTHESIS

The production trade: edge sharpness $\times$ noise $\times$ storage grid



Source: technical note sections 4–8

Desinusoiding changes coordinates—not information

METHOD INVARIANT

$$x(t) = A \sin(2\pi ft + \phi)$$

$$I(t) \longrightarrow I(t(x))$$

- Apply measured position LUT when available
- Correct detector/scanner delay before interpolation
- Preserve direction and invalid turnaround states

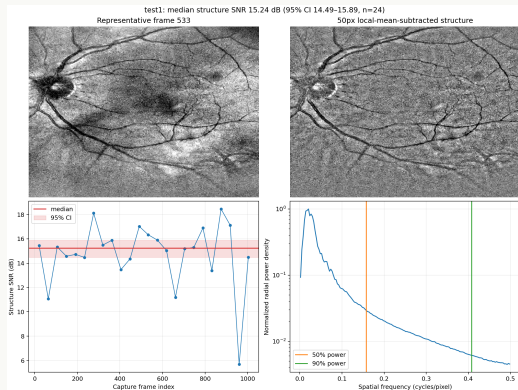
What rectification cannot do

It cannot recover analog bandwidth, photons, or samples that were never acquired. It only maps time-ordered data onto a declared retinal coordinate.

Source: desinusoid.py; scan model; technical note

The image is a time-ordered measurement—not a snapshot

FIRST PRINCIPLES



Source: test1 real retinal capture; measured SNR diagnostic

Raster interpretation

Each stored column/line corresponds to a different acquisition time. Eye motion therefore appears as retinal shear and displacement within a nominal frame.

Required metadata

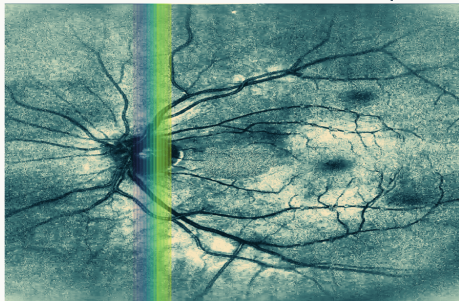
Line start, sample period, direction, position calibration, validity reason, optical state, and hardware clock.

Strip width sets aperture; stride sets output spacing

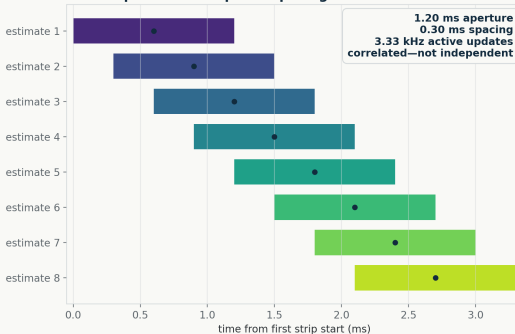
DERIVED + FIXTURE

Overlap buys localization cadence by reusing photons

Width = 24 lines • stride = 6 lines • 75% shared pixels



Aperture and update spacing are different clocks



Source: 20 kline/s candidate clock; 24-line width; 6-line stride

Width and stride control different physical quantities

For strip width W , stride S , and line rate f_L :

$$T_{\text{aperture}} = W/f_L,$$

$$T_{\text{spacing}} = S/f_L,$$

$$f_{\text{output}} = f_L/S,$$

$$\text{shared fraction} = 1 - S/W.$$

With $W = 24$, $S = 6$, and $f_L = 20,000$ lines/s:

$$T_{\text{aperture}} = 1.20 \text{ ms}, \quad T_{\text{spacing}} = 0.30 \text{ ms}.$$

DERIVED EXAMPLE

Interpretation. Each estimate integrates retinal structure over 1.20 ms, yet an estimate is emitted every 0.30 ms. Consecutive estimates share 18 of 24 lines (75%).

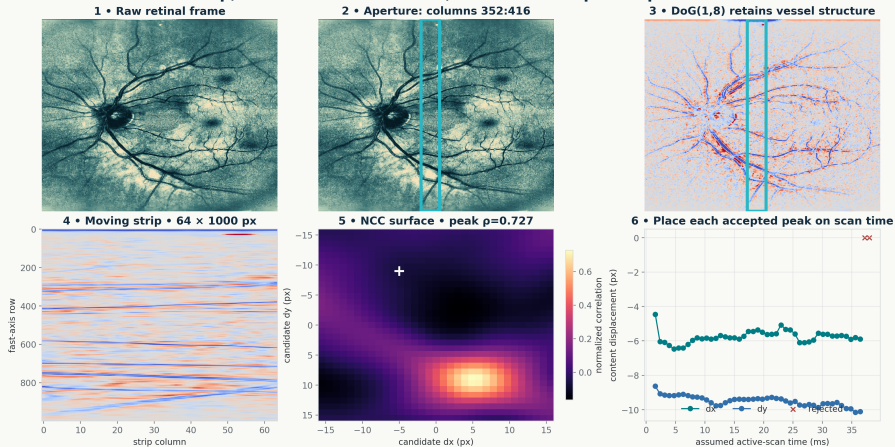
Consequence. The 3.33 kHz output sequence is useful for localizing changes in time, but it is not 3.33 kHz of independent photon evidence. Aperture blur, overlap, detector bandwidth, and registration noise still limit recoverable motion.

Source: technical note candidate line clock; strip-design example in evidence ledger

The registration loop, end to end

IMPLEMENTED METHOD

One strip, one correlation surface, one time-stamped displacement



Source: intraframe_ncc preprocessing, reference, registration, and trajectory modules

Normalized correlation asks for shape, not brightness

METHOD

$$\text{NCC}(u, v) = \frac{\sum_{x,y} [T(x,y) - \bar{T}][R(x+u,y+v) - \bar{R}_{u,v}]}{\sqrt{\sum_{x,y} [T(x,y) - \bar{T}]^2 \sum_{x,y} [R(x+u,y+v) - \bar{R}_{u,v}]^2}}$$

- Local means remove broad brightness offsets
- Normalized energy bounds similarity
- A bounded search region constrains displacement

Preprocessing

Debanding/local-mean removal and DoG filtering expose vessel edges without increasing biological resolution.

Quality boundary

Low, ambiguous, or boundary-saturated peaks remain rejected.

Source: OpenCV `TM_CCOEFF_NORMED`; register.py

How to read a correlation surface

METHOD EXPLANATION

The algorithm shifts the moving strip over a bounded reference search region. At each candidate displacement (u, v) it computes one NCC value. The result is a 2-D surface:

- peak location $\rightarrow$ best displacement;
- peak height $\rightarrow$ similarity/quality;
- peak shape $\rightarrow$ ambiguity and local curvature;
- boundary peak $\rightarrow$ requested motion may exceed the search region.

Subpixel refinement. Integer-pixel matching is refined independently in x and y using a three-point parabolic vertex around the discrete maximum. This estimates the local maximum of the sampled response; it does not create new optical resolution.

Acceptance. A sample is OBSERVED only when acquisition is valid and registration quality passes. Otherwise it is REJECTED or MISSING; prediction is a separate downstream state.

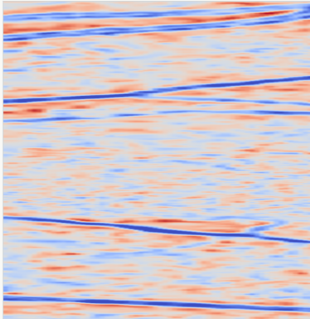
Source: register.py; trajectory.py; production state contract

A correlation peak becomes a placed vessel pattern

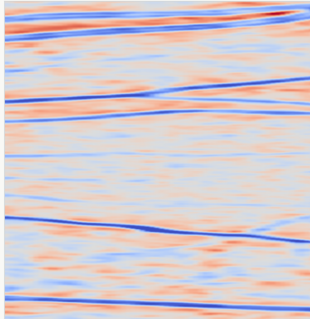
IMPLEMENTED + VISUALIZED

The NCC peak answers: where does this vessel pattern belong?

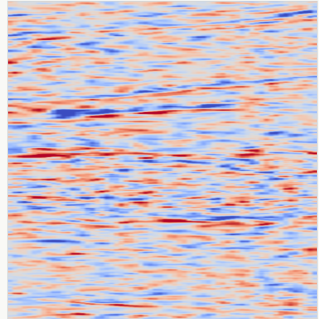
Moving aperture



Peak-selected reference patch



Residual after placement

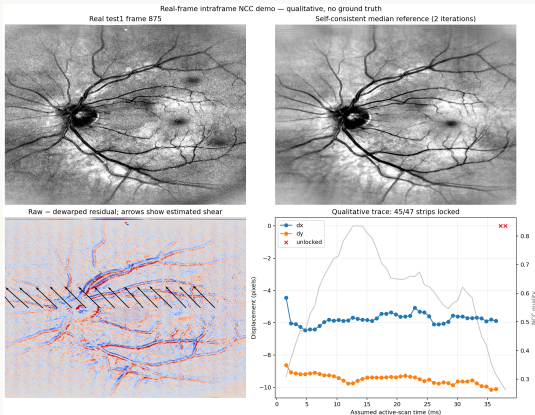


integer placement shown: $dx=-5.00px$, $dy=-9.00px$

Source: integer peak shown for clarity; production uses three-point parabolic refinement

Repeat the placement across scan time to recover a trajectory

MEASURED, UNSCORED



Source: test1 frame 875; intraframe real demonstration

1. Strip center defines acquisition time
2. NCC peak defines displacement
3. Quality defines accepted or rejected
4. States remain explicit in output

Claim boundary

45/47 strips lock in this frame; there is no independent high-rate biological truth.

A trajectory sample has a value, a time, and provenance

DATA CONTRACT

A useful gaze row is more than (x, y) . It should include:

$$z_i = [t_i, \Delta x_i, \Delta y_i, q_i, a_i, e_i, p_i, c_i, r_i],$$

where q_i is registration quality, a_i acquisition state, e_i estimate state, p_i prediction provenance, c_i calibration version, and r_i hardware/software revision.

Why the time is the strip center. The strip integrates several lines; assigning the estimate to the aperture center makes its timing explicit and avoids pretending the displacement was instantaneous.

Why provenance remains attached. Calibration, scanner geometry, optical mode, and estimator settings determine how pixels become physical gaze. Removing them makes later multimodal alignment and validation ambiguous.

Source: `contracts.py`; `trajectory.py`; proposed line-record interface

Pixels become gaze only through explicit calibration

$$\Delta\theta_x = s_x \Delta x_{\text{px}}$$

$$\Delta\theta_y = s_y \Delta y_{\text{px}}$$

- Image axes and scanner axes
- Sign and handedness
- Retinal angle and display coordinates
- Calibration source and version

CALIBRATION REQUIREMENT

Fail loudly

Missing calibration leaves positions in pixels. The pipeline must not invent arcminutes or millimeters.

Separate problem

Axial-length scaling converts angle to retinal distance; it is not part of image registration.

Source: `contracts.py`; calibration configuration; technical note eye model

The reference image is part of the measurement model

METHOD + LIMITATION

INCREMENTAL

High overlap; relative trajectory; accumulated drift.

FIXED / ATLAS

Absolute reference; vulnerable to field-of-view exit.

SELF-CONSISTENT

Improves lock coverage; can absorb shared motion.

A same-acquisition atlas is useful registration structure, but it is not independent gaze truth.

Source: [reference.py](#); self-consistent median reference implementation

Credibility is a ladder; each rung answers a different question

VALIDATION FRAMEWORK

Intended-use cognitive validation

Simultaneous independent human truth

Artificial-eye hardware truth

Synthetic known-reference motion

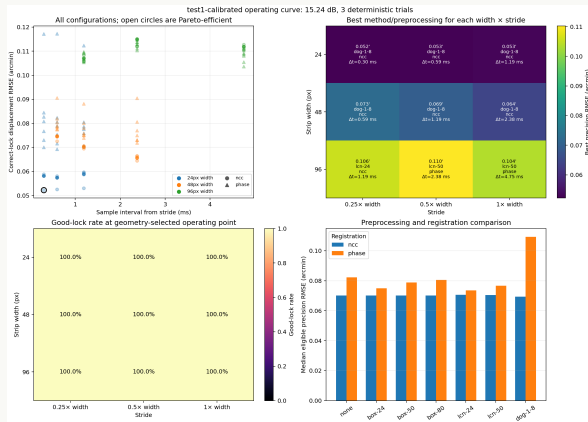
Unit / contract tests

Synthetic precision validates the estimator under its model; it does not establish biological waveform accuracy.

Source: repository tests, synthetic fixtures, and proposed validation program

Measured image statistics set the synthetic operating regime

MEASURED SNR + FIXTURE



Source: snr.json; operating_curve.json; FINDINGS.md

Measured input

Median structure SNR: 15.242 dB
bootstrap interval: 14.485–15.886 dB

Known-truth point

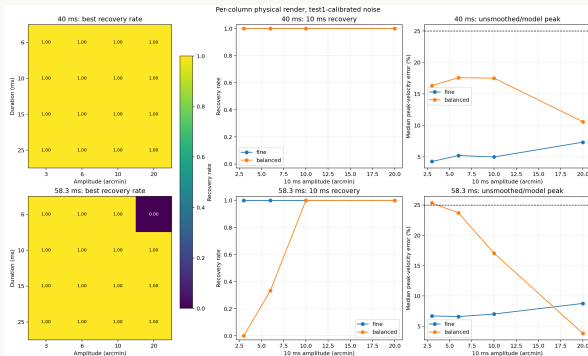
24/6, DoG(1,8), spatial NCC:
0.05221 arcmin RMSE; 393/393
locks

Boundary

Precision is valid only under the
declared synthetic model.

A 10 ms synthetic waveform is recoverable under the stated model

FIXTURE



Fixture

10 arcmin / 10 ms vertical (fast/image-row)
event; real-frame-derived morphology; 15%
speckle + Gaussian noise; 32/4 operating point.

Observed in fixture

3/3 recovered at both 40.0 and 58.3 ms
active-time brackets.

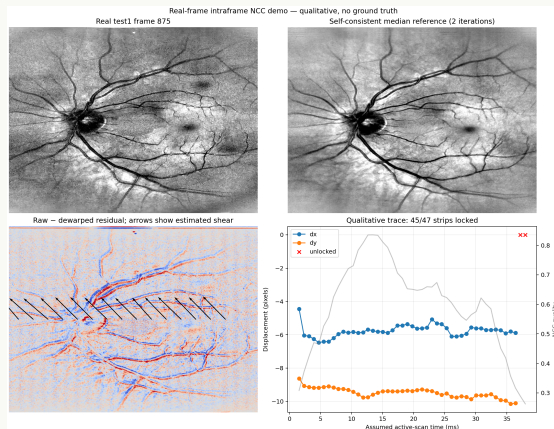
Uncertainty

The 3/3 Wilson interval is 0.439–1.000;
recovery probability remains uncertain.

Source: waveform_recovery.json

Real retinal data demonstrate lock behavior—not truth

MEASURED, UNSCORED



Source: real_demo.json; test1 capture

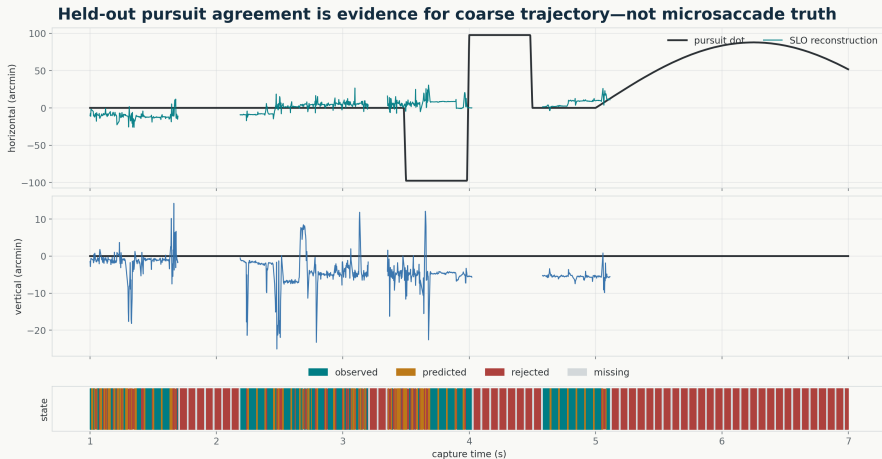
- Frame 875
- 45/47 strips locked
- Median locked NCC 0.6246
- Structured residual and plausible shear

Not established

No independent high-rate eye-motion truth; displacement and waveform accuracy cannot be scored.

The legacy pipeline tracks held-out pursuit at a coarse level

MEASURED + ASSUMED TIMING



Source: champion pursuit result; fixture's unmeasured 10 ms dead-time hypothesis

What the current evidence supports

CLAIM BOUNDARY

SUPPORTED NOW

- Real retinal structure supports strip locking
- Synthetic known-truth registration is precise
- Synthetic 10 ms waveform recovery is demonstrated
- Coarse pursuit correspondence is measured
- OPM coexistence is accepted as characterized

NOT YET SUPPORTED

- Validated human microsaccade waveforms
- Hardware timing inferred from video alone
- Observations inside uncharacterized dead time
- Universal real-capture angular calibration
- Continuous-laser retinal safety determination

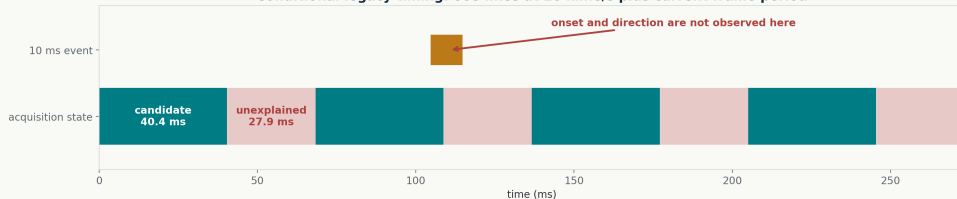
Source: evidence ledger synthesis

The legacy frame boundary may create a cognitive blind interval

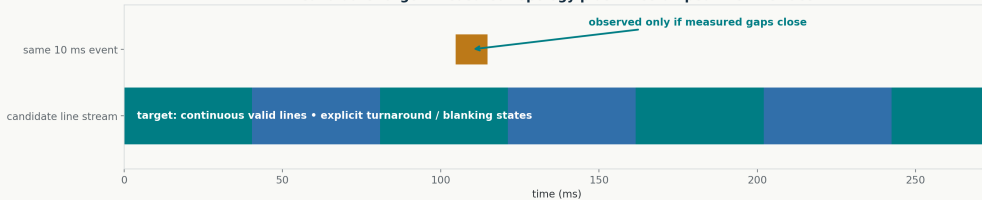
CONDITIONAL DERIVATION

A frame-quality boundary can create a cognitive blind interval

Conditional legacy timing: 808 lines at 20 kline/s plus current frame period



Candidate target: measured topology plus timestamped line interface



Source: 20 kline/s candidate clock; test1 808 lines and 14.633 fps

A good predictor cannot create an observation

METHOD INVARIANT

OBSERVED

Retinal photons acquired; quality gate passed.

PREDICTED

Dynamics model only; uncertainty expands.

REJECTED

Photons acquired; registration not trusted.

MISSING

No usable acquisition; no measurement emitted.

Cognitive event claims require observed photons across the event—or an explicit partial-observation claim.

Source: gap_model.py; contracts.py measurement states

Continuous cognition does not require one fixed image format

CANDIDATE ARCHITECTURE

SCANNER A

Bidirectional raster; best direct SLO continuity.

SCANNER B

Duty-optimized sawtooth; explicit residual turnaround.

FUSION OVERLAY

SLO precision plus eye-camera continuity; pairs with A or B.

Freeze a versioned line record: sequence; clock/time/uncertainty; sample period; direction; position calibration; acquisition state; estimate state; prediction provenance; mode/revisions; optical/power state; intensity.

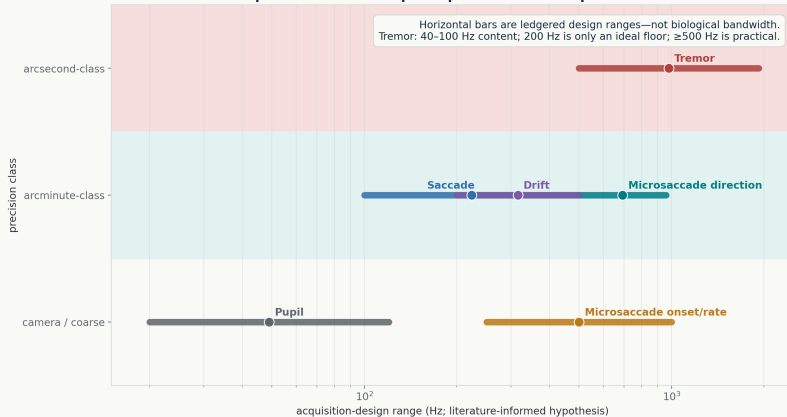
Freeze scanner topology only after measured continuity, exposure, and image-quality tests.

Source: architecture candidates derived from production requirements

Choose sensors on two axes: time and space

LITERATURE SYNTHESIS

Tremor is intrinsically arcsecond-scale; drift and fine direction can become so
Temporal demand and spatial precision are independent axes



Source: literature ranges and acquisition hypotheses in the evidence ledger

Different cognitive signals belong to different sensors

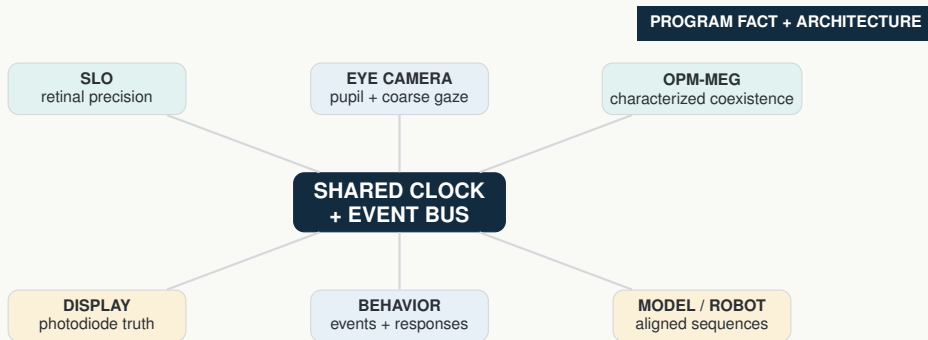
LITERATURE SYNTHESIS

Signal	Scale	Acquisition design	Precision	Primary sensor
Saccade	$> 1-2^\circ$	100–500; ~ 1000 fine onset	arcmin	camera / SLO
Micro onset/rate	$\sim 0.1-0.5^\circ$	≥ 250 ; 500–1000 shape	coarse event	event camera / SLO
Micro direction	endpoint vector	same event samples	sub-arcmin	SLO / fused
Drift	to $\sim 50-94' s^{-1}$	200–500 Hz	$0.5-1'$; arcsec increments	SLO
Tremor	often arcsec	≥ 500 Hz practical	arcsec class	SLO / specialized
Pupil	2–8 mm	20 content; 50–120	0.02–0.05 mm	eye camera

Rates are acquisition targets, not event rates. Tremor has 40–100 Hz content: 200 Hz is only the ideal floor and ≥ 500 Hz is practical. Arcsecond SD-SLO performance remains a validation target.

Source: Rucci & Poletti 2015; Bowers et al. 2019; Rolfs 2009; Kelbsch et al. 2019

The cognitive product is a synchronized sensor graph



OPM coexistence is accepted; production applicability remains bound to a controlled record and revision matrix.

Source: program architecture synthesis

“Laser safety” is three separate regulatory questions

SAFETY PRIMER

1. LASER PRODUCT CLASSIFICATION

IEC 60825-1 and applicable 21 CFR provisions classify accessible emission and require product controls, including scanning safeguards.

2. PATIENT EXPOSURE

ISO 15004-2 or FDA-recognized ANSI Z80.36 evaluate ophthalmic-instrument exposure to the patient's eye under normal and fault conditions.

3. MEDICAL DEVICE PATHWAY

FDA device classification (for example product code MYC) governs regulatory pathway and recognized standards. It is not the same as IEC laser class.

Why scan geometry is insufficient. A 10 kHz scan and nominal spot size describe where and how fast light moves. Exposure also depends on measured spectrum, power at the eye, beam profile, pupil coupling, complete two-axis trajectory and blanking, session duration, repeated visits, tolerances, and scanner-failure behavior.

Source: IEC 60825-1; ISO 15004-2:2024; ANSI Z80.36-2021; FDA MYC and Laser Notice 56

Continuous scanning requires an exposure calculation—not a duty-cycle slogan

PARAMETERIZED ONLY

Local scan contribution

$$t_{\text{dwell}}(x) \approx \frac{d_{\text{eff}}}{v(x)}$$

At scan center, $10 \mu\text{m}/436.8 \text{ m s}^{-1} = 22.9 \text{ ns}$ for the assumed optical diameter.

This is derived geometry, not an exposure limit.

Required standards workflow

1. Classify accessible emission under the applicable IEC/21 CFR product regime.
2. Evaluate patient exposure under the selected ISO/ANSI ophthalmic regime.
3. Apply wavelength, source, aperture, scanning, and temporal corrections within their governing clauses.
4. Analyze scanner stall and other faults as fault conditions with independent controls.

This is not a safety determination: inventing corneal power would produce a false number.

Source: separate IEC/21 CFR product classification from ISO 15004-2 or ANSI Z80.36 patient exposure

The technical note defines geometry; safety still needs source and exposure data

MISSING INPUTS

Continuous acquisition changes exposure duty—but geometry alone cannot establish safety

Known scan geometry

10 kHz full sine

20 kline/s both directions

48° nominal field

10 μm assumed Airy disk

Missing inputs block a safety determination

- × wavelength and spectral bandwidth
- × corneal power and power-control tolerance
- × beam diameter and divergence at the cornea
- × pupil/aperture assumptions required by the applicable standard
- × actual scan waveform, fill pattern, dwell distribution and blanking
- × total exposure duration and repeated-session schedule
- × stationary-beam and scanner-stall fault exposure
- × optical losses and retinal irradiance calibration
- × source angular subtense and extended-source classification
- × applicable standard, edition, classification and medical-device safety review

Allowed output = min(single-pulse rule, average-power rule, repetitive-pulse rule, extended-source correction, and scanner-stall/fault limit) under the selected standard.

Source: cognitive_gaze_evidence.json laser-safety missing-input ledger

Safety belongs in the architecture, not only the verification report

REQUIREMENT

POWER MONITOR

Independent measured optical output.

SCANNER WATCHDOG

Line frequency, amplitude, and phase bounds.

FAIL-SAFE BLANKING

Dark on clock, LUT, or motion invalidity.

EXPOSURE INTEGRATOR

Time-resolved session dose accounting.

HARD FAULT PATH

Independent of application software.

REVISION BINDING

Calculation ↔ optical BOM ↔ firmware.

Continuous tracking is a system-level safety mode: optics, scanner, firmware, interlocks, telemetry, and session logic must be reviewed together.

Source: scanning-source failure modes; IEC 60825-1; 21 CFR 1040.10(f)(9)

Reclassify legacy decisions before production freeze

DECISION LEDGER

KEEP

- Confocal retinal imaging
- High spatial precision
- Calibration discipline
- Safety/interlocks
- Portable mechanics goal

CHANGE

- Frame-centric acquisition
- Uncharacterized dead time
- Filesystem timing
- Offline-only processing
- Single-modality outputs

CONFIGURE

- Field/output grid
- Strip width/stride
- Bandwidth/gain
- Safety-governed modes
- Image/cognition profile

HOLD OPEN

- Final scanner topology
- OCT role in cognition
- Adaptive optics
- AI/VLA product claim
- Continuous optical power

Source: architecture audit synthesis

The no-regret production architecture is line-stream first

CANDIDATE ARCHITECTURE

OPTICS + SENSORS

SLO; OCT option; eye camera; photodiode; OPM triggers

DETERMINISTIC ACQUISITION

ADC; scanner position; line direction; acquisition state; power telemetry

TIME-SYNCD STREAM

Hardware timestamps; clock model; bounded buffers; immutable chunks

REAL-TIME ESTIMATION

Estimate state + prediction provenance; medical image/QC/export; cognitive gaze/events

EXPERIMENT + DATA

Stimulus events; OPM alignment; artifacts; model-ready sequences

Immutable acquisition core → distinct, versioned medical and cognitive product profiles.

Source: scanner, software, cognitive, and production audits

Freeze interfaces before freezing the final hardware topology

REQUIREMENTS

Gate	Metric	Pass boundary
G1 CLOCK	Line timestamp residual	TBD μ S; measured
G2 GEOMETRY	Forward/reverse registration	TBD px; retinal target
G3 CONTINUITY	Observed duty + maximum gap	TBD % / ms
G4 GAZE	Position + onset error	Artificial eye + independent truth
G5 SAFETY	Normal + stall exposure	Formal review pass
G6 OPM	Configuration-bound coexistence	Controlled record + revision matrix
G7 DATA	Sequence-space accounting	Zero silent loss; explicit reasons
G8 LATENCY	Capture-to-feature	TBD percentile budget

A production freeze without quantified intended-use gates only freezes today's assumptions.

Source: proposed measurable production gates; thresholds intentionally not invented

A 30 / 60 / 90-day sequence can retire the highest-risk assumptions

PROPOSAL

0–30 DAYS

- Instrument clocks and validity
- Measure forward/reverse LUTs
- Capture optical source inputs
- Lock intended signal requirements

31–60 DAYS

- Continuous-line prototype
- Artificial-eye motion sweep
- Stall/blanking fault tests
- Display + OPM clock alignment

61–90 DAYS

- Human simultaneous-truth pilot
- Score scanner × fusion options
- Formal exposure review
- Freeze interfaces and gates

Source: proposed validation sequence

Engineering and safety sources

SOURCES

Source	Use
Technical note, 21 Jul 2026	Scanner, bandwidth, Airy, sampling, and output-grid calculations.
ISO 15004-2:2024	International ophthalmic-instrument light-hazard scope.
IEC 60825-1:2014	Laser-product classification, equipment requirements, scanning safeguard.
ANSI Z80.36-2021	FDA-recognized U.S. ophthalmic patient-exposure standard.
FDA product code MYC	Scanning laser ophthalmoscope medical-device classification.
21 CFR 1040.10(f)(9)	Scan-failure protection and accessible-emission inspection.

IEC/21 CFR product classification, ophthalmic patient exposure, and FDA medical-device classification are separate evaluations.

Source: official scope and recognition pages; controlled standards required for compliance

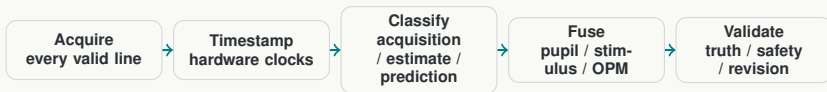
Eye-movement and retinal-scaling sources

SOURCES

Source	Use
Bennett et al. doi:10.1007/BF00175988	1994, Axial-length retinal magnification.
Yang et al. doi:10.1364/OL.40.000085	2015, Sinusoidal rectification and retinal irradiance.
Rucci & Poletti doi:10.1146/annurev-vision-082114-035742	2015, Drift below about 40 Hz; tremor around 40–100 Hz.
Rolfs doi:10.1016/j.visres.2009.08.010	2009, Microsaccade definitions, dynamics, and measurement.
Bowers et al. doi:10.1167/19.11.8	2019, Direct retinal tremor; 1920 Hz sampling, 960 Hz analysis ceiling.
Kelbsch et al. doi:10.3389/fneur.2019.00129	2019, Pupillography bandwidth and acquisition guidance.

Source: full bibliography and evidence use in cognitive_gaze_evidence.json

Keep the retinal precision. Stop letting the frame erase time.



Near-term decision: freeze line-stream, clock, calibration, safety, and multimodal interfaces before freezing production hardware.

Source: engineering evidence and strategic audit